

# Muhammad Umar Khan

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## SUMMARY

M.S. Computer Science student specializing in full-stack and AI-powered systems. Experienced in designing scalable backend architectures, RESTful APIs, and LLM-integrated pipelines for end-to-end applications involving document processing and automation. Proficient in Java, Python, and JavaScript, with a focus on production-grade software engineering.

## EDUCATION

**Wayne State University | Detroit, MI, USA | Jan. 2025 - Aug. 2026**

*M.S. in Computer Science · GPA: 3.89/4*

**University of Windsor | Windsor, ON, Canada | Sep. 2020 – Apr. 2024**

*B.S. in Computer Science with Software Engineering Specialization*

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript/TypeScript, C, HTML, CSS, SQL

**Frameworks & Libraries:** React, Spring Boot, Spring Security, Spring Data JPA (Hibernate), FastAPI, Node.js, Next.js, Vite

**Databases & Tools/DevOps:** PostgreSQL, Docker, SQLite, MongoDB, Supabase, Git, GitHub, JWT Authentication

## EXPERIENCE

### Research Assistant

*University of Windsor | Windsor, ON, Canada | Apr. 2023 - Nov. 2024*

- Engineered a Python-based image segmentation pipeline using clustering to generate super pixels, reducing preprocessing time by ~20% and improving development efficiency
- Evaluated Graph-CNN models for pixel classification, selecting optimal models based on accuracy and runtime
- Modeled and integrated an SQL database with Snowflake and containerized prototypes using Docker for cloud-based development and deployments.

### Teaching Assistant

*University of Windsor | Windsor, ON, Canada | Sep. 2022 – Apr. 2024*

- Conducted weekly office hours and graded assignments/exams for 100+ students enrolled in Data Structures & Algorithms, Database Management Systems, and Object-Oriented Programming (Java).
- Assisted lab sessions and exams, and aided in reinforcing core concepts in algorithms, database systems, and OOP

## PROJECTS

**AI Resume Tailoring Platform | React, Vite, FastAPI, Python, Google Gemini API, pdfplumber, BeautifulSoup, ReportLab**

- Engineered a full-stack AI resume tailoring platform (FastAPI + React) with a single-pass Gemini pipeline for simultaneous keyword extraction and ATS rewriting, cutting API rate-limit exposure by 50% and improving development efficiency
- Built a real-time SSE streaming pipeline emitting live backend progress events at each processing stage, replacing a fake progress timer with accurate inference-time feedback streamed to the frontend to improve system visibility.
- Designed a 5-metric ATS scoring engine (parsability, keyword injection, coverage, quantification rate, job match) that programmatically scans tailored resume text and surfaces actionable improvement insights per generation

**Market Predictor | React, Next.js, Supabase, Google OAuth, Recharts, JavaScript**

- Engineered a full-stack market analysis platform with Google OAuth authentication via Supabase and secure server-side API proxying to protect third-party keys and ensure reliability.
- Architected a modular 20-component React application with shared utilities, real-time price alert polling, and responsive mobile-first design

**Blog Platform | Java, Spring Boot, Spring Security, JWT, React, TypeScript, Vite, Tailwind CSS, PostgreSQL**

- Developed a secure full-stack application with JWT-based authentication and role-based access control using Java Spring Boot and Spring Security to maintain secure software.
- Designed RESTful APIs following a layered architecture (Controller → Service → Repository) and integrated a relational PostgreSQL schema to connect application components with databases.
- Implemented secure authentication workflows including password hashing, token-based sessions, and protected routes to maintain secure, consistent, and maintainable software.